

Mathematical Logic

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Spring 2026

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1 Propositional logic

1.1 Propositional formulas

Definition 1.1.1. A proposition is a statement which is either **true** (T) or **false** (F). Given propositional variables p and q we can combine these into longer propositions using the following connective, which are represented by truth tables:

p	q	$(\neg p)$	$(p \wedge q)$	$(p \vee q)$	$(p \rightarrow q)$	$(p \leftrightarrow q)$	$(p \downarrow q)$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	F
F	T	T	F	T	T	F	F
F	F	T	F	F	F	T	T

All **propositional formulas** are either:

- propositional variables like p or q ;
- the negation $(\neg\phi)$ for some propositional formula ϕ ;

- a connective applied to two propositional formulas $(\phi \cdot \psi)$ for $\cdot \in \{\wedge, \vee, \rightarrow, \leftrightarrow, \downarrow\}$.

Because of the strict bracketing, we can use induction to prove every formula is either a propositional variable or build from strictly shorter formulas in a unique way.

Lemma 1.1.2. The connectives $\vee$ and $\wedge$ are associative, so we often omit brackets in expressions like $(p_1 \vee (p_2 \vee p_3))$ for $(p_1 \vee p_2 \vee p_3)$.

Proof. Truth tables, and induction on number of connectives. $\square$

Definition 1.1.3. For $n \in \mathbb{N}$, a **truth function** of n variables is a function $f : \{T, F\}^n \rightarrow \{T, F\}$. Given a propositional formula ϕ with variables among $\{p_1, \dots, p_n\}$, there is a associated truth function F_ϕ whose value at $(x_1, \dots, x_n)$ is the truth value of ϕ when each p_i has value x_i .

Lemma 1.1.4. There are 2^{2^n} truth functions of n variables.

Proof. These are just functions, so we can count. $\square$

Definition 1.1.5. A set of connectives S is called **adequate** if for all $n \geq 1$, for all truth functions f of n variables, there exists some formula which only involves connectives from S , and variables $p_1, \dots, p_n$.

Theorem 1.1.6. The set $\{\neg, \wedge, \vee\}$ is adequate.

Proof. Let $G : \{T, F\}^n \rightarrow \{T, F\}$ be the truth function. If $G \equiv F$ on all inputs, then take $\phi = (p_1 \wedge (\neg p_1))$, and we have $G = F_\phi$.

Now write $\{v_1, \dots, v_k\}$ for the set of all $v \in \{T, F\}^n$ such that $G(v) = T$. For each $v_i = (v_{i1}, \dots, v_{in})$ we define:

$$q_{ij} = \begin{cases} p_j & \text{if } v_{ij} = T \\ (\neg p_j) & \text{if } v_{ij} = F \end{cases}$$

and then define $\psi_i := (q_{i1} \wedge \dots \wedge q_{in})$, a propositional formula which is true iff each p_j has the same value as v_{ij} . Now take $\phi = (\psi_1 \vee \dots \vee \psi_k)$, so we have $G = F_\phi$, where ϕ only consists of connectives in $\{\neg, \wedge, \vee\}$. $\square$

The form of propositions used in this proof is **disjoint normal form** written dnf.

Corollary 1.1.7. Suppose χ is a formula whose truth function is not always F , then χ is logically equivalent to a formula in disjunctive normal form.

Corollary 1.1.8. To show any other set of connectives S is adequate, we just have to construct the truth tables for $\{\neg, \wedge, \vee\}$ from connectives in S .

Corollary 1.1.9. The sets of connectives $\{\neg, \vee\}$, $\{\neg, \wedge\}$, $\{\neg, \rightarrow\}$, and $\{\downarrow\}$ are all adequate.

Definition 1.1.10. A propositional formula ϕ is called a **tautology** if its truth function always has value T . If ψ is another propositional formula, then we say ϕ and ψ are **logically equivalent** iff $(\phi \leftrightarrow \psi)$ is a tautology.

1.2 The formal system K

Definition 1.2.1. The **formal system** L (for propositional logic) has the following ingredients:

- An **alphabet** of symbols which consists of **variables** $\{p_1, p_2, \dots\}$, **connectives** $\{\neg, \rightarrow\}$, and **punctuation** $\{(\, , \,)\}$.
- **Formulas**, which are certain finite sequences of symbols from the alphabet constructed as follows
 - any variable is a formula;
 - If ϕ and ψ are formulas of L , then so are $(\neg\phi)$ and $(\phi \rightarrow \psi)$
 - All formulas arise in this way.

We call these L -**formulas**.

- **Axioms**, suppose ϕ, ψ, χ are L -formulas, then the following are **Axioms of L** :

- (A1) $(\phi \rightarrow (\psi \rightarrow \phi))$
- (A2) $((\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi)))$
- (A3) $((\neg\psi) \rightarrow (\neg\phi)) \rightarrow (\phi \rightarrow \psi)$

• **Deduction rules**, in L this is just modus ponens

(MP) From ϕ , and $(\phi \rightarrow \psi)$, deduce ψ .

The collection of an alphabet, formulas, axioms, and deduction rules, constitute a general **formal deduction system**.

Definition 1.2.2. A **proof** in a formal deduction system Σ is a finite sequence of Σ -formulas, each of which is either an axioms, or deduced from previous formulas in the proof by a deduction rule. The last formula in a proof is called a **theorem** of Σ . The number of formulas in the proof is called its **length**.

The main goal of this section is to show that the theorems of L are precisely the tautologies.

Definition 1.2.3. Suppose Γ is a set of L -formulas, then a **deduction from** Γ is a finite sequence of L -formulas, each of which is either an axioms, in Γ , or a deduction from the previous terms. Write $\Gamma \vdash_L \phi$ if there is a deduction from Γ ending in ϕ , in which case we say ϕ is a **consequence** of Γ .

The theorems of L are precisely the formulas ϕ such that $\emptyset \vdash_L \phi$, we normally write this as $\vdash_L \phi$.

Proposition 1.2.4 (Deduction theorem). Suppose Γ is a set of L -formulas and ϕ, ψ are L -formulas such that $\Gamma \cup \{\phi\} \vdash_L \psi$, then $\Gamma \vdash_L (\phi \rightarrow \psi)$.

Proof. We go by induction on n , the length of the deduction $\Gamma \cup \{\phi\} \vdash_L \psi$.

If $n = 1$, then ψ is either an axiom, in Γ , or is ϕ . In the first two cases $\Gamma \vdash_L \psi$ is a one line deduction. Otherwise, showing $(\phi \rightarrow \phi)$ is a theorem is a classic exercise.

For the inductive step, in the deduction $\Gamma \cup \{\phi\} \vdash_L \psi$, then final line is either ψ , in Γ , or is ϕ , in which case we argue as in the inductive step. Otherwise ψ is obtained from earlier steps using MP, in which case there are formulas χ and $(\chi \rightarrow \psi)$ earlier on in the deduce, we apply the induced to step to get $\Gamma \vdash_L (\phi \rightarrow \chi)$ and $\Gamma \vdash_L (\phi \rightarrow (\chi \rightarrow \psi))$. Now (A2) and two uses of MP, gives $\Gamma \vdash_L (\phi \rightarrow \psi)$ as required. $\square$

Corollary 1.2.5 (Hypothetic syllogism). Suppose ϕ, ψ, χ are L -formulas, such that $\vdash_L (\phi \rightarrow \psi)$ and $\vdash_L (\psi \rightarrow \chi)$, then $\vdash_L (\phi \rightarrow \chi)$.

Proof. We use the deduction theorem for $\Gamma = \emptyset$ on $\vdash_L (\phi \rightarrow \psi)$, and modus ponenes on the two theorems we have to deduce χ , now use the converse of the deduction theorem to deduce $\phi \rightarrow \chi$. $\square$

I will frequently write just $\vdash$ for $\vdash_L$.

1.3 Soundness

Definition 1.3.1. A **propositional valuation** is an assignment of truth values to the propositional variable $\{p_1, p_2, \dots\}$ which we can view as a function $v : \mathbb{N} \rightarrow \{T, F\}$. Using truth tables, we can extend v to a function on the set of L -formulas. This will have some immediate properties such as $v((\neg\phi)) \neq v(\phi)$ and $v((\phi \rightarrow \psi)) = F$ iff $v(\phi) = T$ and $v(\psi) = F$.

Valuations correspond to rows in a truth table.

Theorem 1.3.2 (Soundness of L). Suppose ϕ is a theorem of L , then ϕ is a tautology.

Proof. We do induction on the length of a proof of $\vdash \phi$.

In the base case, ϕ must be an axiom, by truth tables three axioms are tautologies.

In the inductive step, we assume ϕ has a proof of length n , the last step is either: an axiom, in which case we argue as in the base case; or modus ponens from two theorems χ and $(\chi \rightarrow \phi)$, by induction we know both of these are tautologies, and now by considering the truth tables, we know χ and $(\chi \rightarrow \phi)$ are both true implies ϕ is true. $\square$

Using the same proof strategy, there is an immediate generalisation.

Corollary 1.3.3 (Generalisation of soundness). Suppose Γ is a set of L formulas and ϕ is an L -formula such that $\Gamma \vdash \phi$. Suppose v is a valuation with $v(\Gamma) = \{T\}$, then $v(\phi) = T$.

1.4 Completeness

Definition 1.4.1. Suppose Γ is a set of L -formulas. We say Γ is **consistent** if there is no L -formula ϕ such that $\Gamma \vdash \phi$ and $\Gamma \vdash (\neg\phi)$. We say Γ is **complete** if for every L -formula ϕ either $\Gamma \vdash \phi$ or $\Gamma \vdash (\neg\phi)$.

Unfortunately, the completeness theorem says in the case $\Gamma = \emptyset$, Γ is consistent. But in this case, Γ is not complete. The word complete is being used with two *completely* different meanings.

Proposition 1.4.2. Suppose Γ is a consistent set of L -formulas and $\Gamma \not\vdash \phi$, then $\Gamma \cup \{(\neg\phi)\}$ is consistent.

Proof. Assume for a contradiction there exists some L -formula ψ such that

$$\Gamma \cup \{(\neg\phi)\} \vdash \psi, (\neg\psi)$$

by passing the second statement through the deduction formula we get $\Gamma \vdash ((\neg\phi) \rightarrow (\neg\psi))$, we then use (A3) to get $\Gamma \vdash (\psi \rightarrow \phi)$, we can now use the first statement to get $\Gamma \cup \{(\neg\phi)\} \vdash \phi$. Note that for any L -formula χ , the formula $((\neg\chi) \rightarrow \chi) \rightarrow \chi$ is a theorem of L , so we get $\Gamma \vdash \phi$, which contradicts the assumption, thus $\Gamma \cup \{(\neg\psi)\}$ is consistent. $\square$

Proposition 1.4.3 (Lindenbaum lemma). Suppose Γ is a consistent set of L -formulas. Then there is a set of L -formulas $\Gamma^* \supset \Gamma$ which is consistent and complete.

Proof. The set of L -formulas is countable, so we can list them as $\phi_0, \phi_1, \phi_2, \dots$ and inductively define a chain of sets of formulas:

$$\Gamma_0 \subseteq \Gamma_1 \subseteq \dots, \quad \Gamma_0 = \Gamma, \quad \Gamma_{n+1} = \begin{cases} \Gamma_n & \text{if } \Gamma_n \vdash \phi_n \\ \Gamma_n \cup \{(\neg\phi_n)\} & \text{if } \Gamma_n \not\vdash \phi_n \end{cases}$$

Now by induction with the previous proposition, each of the Γ_i is consistent. We let

$$\Gamma^* = \bigcup_{i \in \mathbb{N}} \Gamma^i$$

and claim Γ^* is consistent. If $\Gamma^* \vdash \phi, (\neg\phi)$, then as deduction have finite length, there exists some n with $\Gamma_n \vdash \phi, (\neg\phi)$, a contradiction. To show Γ^* is complete, suppose χ is such that $\Gamma^* \not\vdash \chi$, there is some n with $\phi_n = \chi$ and so $\Gamma_n \not\vdash \chi$, thus $(\neg\chi) \in \Gamma_{n+1}$, and therefore $\Gamma^* \vdash (\neg\chi)$. $\square$

Lemma 1.4.4. Suppose Γ^* is a complete, consistent set of L -formulas. Then there is a valuation v such that, for every L -formula ϕ , $v(\phi) = T$ iff $\Gamma^* \vdash \phi$.

Proof. As Γ^* is complete, we know for every propositional variable p_i , either $\Gamma^* \vdash p_i$ or $\Gamma^* \vdash (\neg p_i)$. Let v be the unique valuation such that $v(p_i) = T$ iff $\Gamma^* \vdash p_i$. We will now do induction on the length of a formula ϕ , to show v satisfies our desired property, in the base case this is by definition. For the inductive step, there are two possible cases.

Case 1: ϕ is $(\neg\psi)$ for some ψ . Here, $v(\phi) = T$ iff $v(\psi) = F$ iff $\Gamma^* \not\vdash \psi$ iff $\Gamma^* \vdash (\neg\psi)$ iff $\Gamma^* \vdash \phi$.

Case 2: ϕ is $(\psi \rightarrow \chi)$. Here, $v(\phi) = F$ iff $v(\psi) = T$ and $v(\chi) = F$ iff $\Gamma^* \vdash \psi$ and $\Gamma^* \not\vdash \chi$. If $\Gamma^* \vdash \phi$ then we could do modus ponens to get $\Gamma^* \vdash \chi$ which contradicts consistency. If $\Gamma^* \not\vdash \phi$ then $\Gamma^* \not\vdash \chi$, otherwise by (A1) we get $\vdash (\chi \rightarrow (\psi \rightarrow \chi))$; similarly if $\Gamma^* \vdash (\neg\psi)$, then by the problem sheet we know $\vdash ((\neg\psi) \rightarrow (\psi \rightarrow \chi))$, thus by the inductive hypothesis $v(\chi) = F$ and $v((\neg\psi)) = F$, thus $v(\phi) = F$. $\square$

Corollary 1.4.5. Suppose Γ is a consistent set of L -formulas, then there is a valuation v with $v(\Gamma) = T$.

Corollary 1.4.6. Suppose Δ is a set of L -formulas which is consistent and $\Delta \not\vdash \phi$, then there is a valuation with $v(\Delta) = T$ and $v(\phi) = F$.

Theorem 1.4.7 (Generalised completeness of L). If Δ is a consistent set of L -formulas, and ϕ is an L -formula such that for every valuation v with $v(\Delta) = T$, we have $v(\phi) = T$, then $\Delta \vdash \phi$.

Proof. This is the contrapositive of the previous corollary. $\square$

Corollary 1.4.8 (Completeness of L). If the L -formula ϕ is a tautology, then $\vdash \phi$.

Theorem 1.4.9 (Compactness theorem for L). Suppose Γ is a set of L -formulas, then tfae:

1. There is a valuation v with $v(\Gamma) = T$

2. For every finite subset $\Sigma \subseteq \Gamma$, there is a valuation w with $w(\Sigma) = T$.

Proof sketch. (1) definitely implies (2), so suppose (1) doesn't hold. By the first corollary, Γ isn't consistent. As deductions are finite, there is a finite subset $\Sigma \subseteq \Gamma$ which is also inconsistent, then by the generalised soundness theorem, the desired valuation of (2) cannot exist. $\square$

2 First-order logic

2.1 Structures

Definition 2.1.1. Suppose A is a set and $n \geq 1$: an **n -ary relation** on A is a subset $\bar{R} \subseteq A^n$; an **n -ary function** on A is a function $\bar{f} : A^n \rightarrow A$.

Some examples are: equality and orders as 2-ary relationships, various binary operations as 2-ary functions, subsets as 1-ary relations. In future for $\bar{R} \subseteq A^n$ and $(a_1, \dots, a_n) \in A^n$ we write $\bar{R}(a_1, \dots, a_n)$ for $(a_1, \dots, a_n) \in \bar{R}$.

Definition 2.1.2. A **first-order structure** $\mathcal{A}$ consists of:

- a nonempty set A , called the **domain** of $\mathcal{A}$;
- a set $\{\bar{R}_i \mid i \in I\}$ of **relations** on A , $\bar{R}_i \subseteq A^{n_i}$ for $i \in I$;
- a set $\{\bar{f}_j \mid j \in J\}$ of **functions** $\bar{f}_j : A^{m_j} \rightarrow A$ for $j \in J$;
- a set $\{\bar{c}_k \mid k \in K\} \subseteq A$ of **constants**.

The **indexing sets** I, J, K can be empty, and are normally subsets of $\mathbb{N}$. The information:

$$(n_i \mid i \in I) \quad (m_j \mid j \in J) \quad K$$

is called the **signature** of $\mathcal{A}$. We will often write

$$\mathcal{A} = \langle A; (\bar{R}_i \mid i \in I), (\bar{f}_j \mid j \in J), (\bar{c}_k \mid k \in K) \rangle$$

Now for some more fleshed out examples:

- Any ordered set A (e.g. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$) can be a first order structure with $I = \{1\}$, $J = K = \emptyset$ and $\bar{R}_1(a_1, a_2)$ to mean $a_1 < a_2$.
- Any group G with a binary relation $\bar{R}$ for equality, a binary function $\bar{m}$ for multiplication, a unary function $\bar{i}$ for inversion, and the constant element $\bar{e}$.
- Any ring R with a binary relation $\bar{R}$ for equality, two binary functions $\bar{m}$ and $\bar{e}$ for multiplication and addition respectively, the unary function $\bar{n}$ for negation, and the constants $\bar{0}$ and $\bar{1}$.
- Peano arithmetic $\langle \mathbb{N}; \bar{R}, \bar{m}, \bar{a}, \bar{s}, \bar{0} \rangle$ where $\bar{R}, \bar{m}, \bar{a}, \bar{0}$ are as in a ring, and $\bar{s}$ is the successor function $\bar{s}(n) = n + 1$.
- Graphs with two binary relations $\bar{R}$ for equality and $\bar{E}$ for adjacency.

Definition 2.1.3. A **first-order language** $\mathcal{L}$ has an alphabet of symbols:

variables	$x_0, x_1, \dots$
connectives	$\neg, \rightarrow$
punctuation	$(,)$
quantifier	$\forall$
relation symbols	R_i for $i \in I$
function symbols	f_j for $j \in J$
constant symbols	c_k for $k \in K$

where I, J, K are indexing sets, which could be empty. Each symbol R_i or f_j comes equipped with an arity n_i or m_j respectively, and the signature of $\mathcal{L}$ can be defined as with structures.

An **$\mathcal{L}$ -structure** is a structure $\mathcal{A}$ with the same signature as $\mathcal{L}$, there is a correspondence between relation, function, and constant symbols of $\mathcal{L}$ with the actual relations, functions, and constants of $\mathcal{A}$ such that arities agree. This correspondence is called an **interpretation** of $\mathcal{L}$.

In our previous examples, we have given the first-order languages of groups, rings, etc where the actual mathematical objects are exactly the $\mathcal{L}$ -structure for these languages.

Definition 2.1.4. A **term** of $\mathcal{L}$ is defined as follows:

- any variable is a term;
- any constant symbol is a term;
- if f is an m -ary function symbol of $\mathcal{L}$ and $t_1, \dots, t_m$ are terms, then $f(t_1, \dots, t_m)$ is also a term;
- all terms arise in this way.

An **atomic formula** of $\mathcal{L}$ is of the form $R(t_1, \dots, t_n)$ where R is an n -ary relation symbol of $\mathcal{L}$ and $t_1, \dots, t_n$ are terms.

An **$\mathcal{L}$ -formula** is defined as follows:

- an atomic formula is an $\mathcal{L}$ -formula;
- If ϕ and ψ are $\mathcal{L}$ -formulas, then $(\neg\phi)$, $(\phi \rightarrow \psi)$, and $(\forall x)\phi$ are all $\mathcal{L}$ -formulas, where x is any variable;
- every $\mathcal{L}$ -formula arises in this way.

Definition 2.1.5. With the same notation as above, if $\mathcal{A}$ is an $\mathcal{L}$ -structure, a **valuation** in $\mathcal{A}$ is a function v from the set of terms of $\mathcal{L}$ to A such that for all $k \in K$, $v(c_k) = \bar{c}_k$; and for all terms $t_1, \dots, t_m$ and m -ary function symbols f , $v(f(t_1, \dots, t_m)) = \bar{f}(v(t_1), \dots, v(t_m))$.

Lemma 2.1.6. Suppose $\mathcal{A}$ is an $\mathcal{L}$ -structure and $a_0, a_1, \dots \in A$, then there is a unique valuation v in $\mathcal{A}$ with $v(x_l) = a_l$ for all $l \in \mathbb{N}$.

Proof. By induction on the length of the terms, if we make v satisfy everything, then we can build v for all of the terms. $\square$

Definition 2.1.7. Suppose $\mathcal{A}$ is an $\mathcal{L}$ -structure, and x_l is a variable. If v, w are valuations in $\mathcal{A}$ such that for all $m \neq l$, $v(x_m) = w(x_m)$, then we say v and w are x_l -**equivalent**.

Definition 2.1.8. Suppose $\mathcal{A}$ is an $\mathcal{L}$ -structure, and v is a valuation in $\mathcal{A}$, then for all $\mathcal{L}$ -formulas ϕ , define recursively v **satisfies** ϕ , written $v[\phi] = T$, in $\mathcal{A}$ if:

- if R is an n -ary relation symbol and $t_1, \dots, t_n$ are terms of $\mathcal{L}$, then v satisfies $R(t_1, \dots, t_n)$ iff $\bar{R}(v(t_1), \dots, v(t_n))$ holds in $\mathcal{A}$;
- if ϕ and ψ are $\mathcal{L}$ -formulas and we know if v satisfies these, then:
 - v satisfies $(\neg\phi)$ in $\mathcal{A}$ iff v does not satisfy ϕ in $\mathcal{A}$,
 - v does not satisfy $(\phi \rightarrow \psi)$ in $\mathcal{A}$ iff v satisfies ϕ in $\mathcal{A}$ but v doesn't satisfy ψ in $\mathcal{A}$,
 - v satisfies $(\forall x_l)\phi$ in $\mathcal{A}$ iff for all valuations w in $\mathcal{A}$ such that w is x_l -equivalent to v , then w satisfies ϕ in $\mathcal{A}$.

If every valuation in $\mathcal{A}$ satisfies the $\mathcal{L}$ -formula ϕ we say ϕ is **true** in $\mathcal{A}$ or $\mathcal{A}$ is a **model** of ϕ and write $\mathcal{A} \models \phi$. If $\mathcal{A} \models \phi$ for every $\mathcal{L}$ -structure $\mathcal{A}$, we say ϕ is **logically valid** and write $\models \phi$.

Logically valid formulas are clearly the tautologies of first-order languages. While there is an algorithm, by truth tables, to determine if any propositional formula is a tautology, Gödel's incompleteness theorem tells us there is no such algorithm for a general $\mathcal{L}$ -formula.

Definition 2.1.9. Suppose ϕ and ψ are $\mathcal{L}$ -formulas, then we will use the notation $(\exists x)\phi$ for $(\neg(\forall x)(\neg\phi))$ and $(\phi \vee \psi)$ for $((\neg\phi) \rightarrow \psi)$ and the other propositional logic symbols.

Lemma 2.1.10. Suppose $\mathcal{A}$ is an $\mathcal{L}$ -structure and ϕ is an $\mathcal{L}$ -formula. Let v be a valuation in $\mathcal{A}$, then v satisfies $(\exists x_1)\phi$ in $\mathcal{A}$ iff there is a valuation w , which is x_1 -equivalent to v , such that w satisfies ϕ .

Proof. Suppose v satisfies $(\neg(\forall x_1)(\neg\phi))$, then v does not satisfy $(\forall x_1)(\neg\phi)$. So there is a valuation w , x_1 -equivalent to v , such that w does not satisfy $(\neg\phi)$. Such a valuation w satisfies ϕ . $\square$

Definition 2.1.11. Suppose χ is an $\mathcal{L}$ -formula involving the propositional variables $p_1, \dots, p_n$. Suppose $\mathcal{L}$ is a first-order language and $\phi_1, \dots, \phi_n$ are $\mathcal{L}$ -formulas. A **substitution instance** of χ is obtained by replacing each p_i in χ by ϕ_i , call the resulting $\mathcal{L}$ -formula θ .

Theorem 2.1.12. With the above notation, we have the following:

1. θ is an $\mathcal{L}$ -formula;
2. Suppose v is a valuation in an $\mathcal{L}$ -structure $\mathcal{A}$, and w is the propositional valuation with $w(p_i) = v[\phi_i]$ for all i , then $v[\theta] = w(\chi)$;
3. if χ is a tautology, then θ is logically valid.

Proof. We omit the proof of (1), as (3) follows from (2), proving (2) is all that remains, and this follows exactly from induction on the length of χ . $\square$

There are many logically valid formulas which don't arise in this way.

2.2 Bound and free variables

Definition 2.2.1. suppose ϕ and ψ are $\mathcal{L}$ -formulas and $(\forall x_i)\phi$ occurs as a subformula of ψ , then we say

- ϕ is in the **scope** of the quantifier $(\forall x_i)$. An occurrence of a variable x_j in ψ is **bound** if it is in the scope of a quantifier $(\forall x_j)$ in ψ .
- Otherwise, it is a **free** occurrence of x_j , variables have a free occurrence in ψ are called the **free variables** of ψ .
- A formula with no free variables is called a **closed formula** or a **sentence** of $\mathcal{L}$.

Definition 2.2.2. If ψ is an $\mathcal{L}$ -formula with free variables amongst $x_1, \dots, x_n$, then we may write $\psi(x_1, \dots, x_n)$ for ψ . If $t_1, \dots, t_n$ are terms, then by $\psi(t_1, \dots, t_n)$ we mean the $\mathcal{L}$ -formula obtained by replacing each **free** occurrence of x_i in ψ by t_i .

Proposition 2.2.3. Suppose ϕ is a closed $\mathcal{L}$ -formula and $\mathcal{A}$ is an $\mathcal{L}$ -structure, then either $\mathcal{A} \models \phi$ or $\mathcal{A} \models (\neg\phi)$. More generally, if ϕ has free variables amongst $x_1, \dots, x_n$ and v, w are valuations in $\mathcal{A}$ with $v(x_i) = w(x_i)$ for all i , then $v[\phi] = T \iff w[\phi] = T$.

Proof. Note that the first statement follows from the generalisation with $n = 0$. We will do induction on the number of connectives and quantifiers in ϕ .

For the base case ϕ is atomic, so we let $\phi = R(t_1, \dots, t_m)$ where there t_j are terms. The t_j only involve variables amongst $x_1, \dots, x_n$, now as v and w agree on all these, we will have

$$v[R(t_1, \dots, t_m)] = T \iff \bar{R}(v(t_1), \dots, v(t_m)) \iff \bar{R}(w(t_1), \dots, w(t_m)) \iff w[R(t_1, \dots, t_m)] = T$$

For the inductive step, ϕ is either $(\neg\psi)$, $(\psi \rightarrow \chi)$ or $(\forall x_i)\psi$, the first two cases are left as exercises, we will do the third. If $v[\phi] = F$, then there is a valuation v' which is x_i -equivalent to v with $v'[\psi] = F$. Let w' be the valuation x_i -equivalent to w with $w'(x_i) = v'(x_i)$, then v' and w' agree on the free variables $x_1, \dots, x_n$ of ψ , so by the inductive hypothesis we get $v'[\psi] = w'[\psi]$. So if $w'[\psi] = F$, and thus $w[\phi] = F$. $\square$

If $\mathcal{A}$ is an $\mathcal{L}$ -structure and $\psi(x_1, \dots, x_n)$ is an $\mathcal{L}$ -formula, with $a_1, \dots, a_n \in A$, the domain of $\mathcal{A}$, then we write $\mathcal{A} \models \psi(a_1, \dots, a_n)$ to mean $v[\psi] = T$ for every valuation v in $\mathcal{A}$ with $v(x_i) = a_i$ for all i .

2.3 Substituting terms for variables

Definition 2.3.1. Let ϕ be an $\mathcal{L}$ -formula, x_i a variable, and t an $\mathcal{L}$ -term, we say t is **free for x_i in ϕ** if there is no variable x_j in t such that x_i has a free occurrence within the scope of a quantifier $(\forall x_j)$ in ϕ .

Theorem 2.3.2. Suppose $\phi(x_1)$ is an $\mathcal{L}$ -formula, possible with other free variables, and let t be a term free for x_1 in ϕ , then $\models ((\forall x_1)\phi(x_1) \rightarrow \phi(t))$.

Proof. As per usual, we will go by induction on the number of connectives and quantifiers in ϕ .

For the base case, ϕ is an atomic formula $R(u_1, \dots, u_m)$. Let u_i^* be the result of substituting t for x_1 in u_i . Then by induction on the length of the terms, each u_i^* is a term, and $v'(u_i) = v(u_i^*)$ $\square$

2.4 Comparing structures

2.5 The formal system $K_{\mathcal{L}}$

2.6 Gödel's completeness theorem

2.7 Equality

2.8 Models

3 Set theory

3.1 Elementary cardinality

Definition 3.1.1. The sets A, B are **equinumerous** (or **have the same cardinality**) if there exists a bijection $f : A \rightarrow B$, we write this as $A \approx B$.

Definition 3.1.2. A set A is **finite** (resp. **countable infinite**) if it is equinumerous with some natural number (resp. $\mathbb{N}$), and just **countable** if it is either of those.

Recall the following basic facts about countable sets from IUM:

- Every subset of a countable set is countable;
- a set A is countable iff there exists an injective function $f : A \rightarrow \mathbb{N}$;
- if A and B are countable then $A \times B$ is countable;
- if $A_0, A_1, \dots$ are all countable, then their union $\bigcup_n A_n$ is also countable, but this uses the axiom of choice.

Theorem 3.1.3 (Cantor). For all sets X , there is no surjective function $f : X \rightarrow \mathcal{P}(X)$.

Proof. Suppose, for a contradiction, there exists such an f , then consider the set

$$Y = \{y \in X \mid y \notin f(y)\}$$

As f is surjective, there exists a set $z \in X$ with $f(z) = Y$, but now if we assume either $z \in Y$ or $z \notin Y$ we get a contradiction. $\square$

Definition 3.1.4. For sets A, B , write $|A| \leq |B|$ if there exists an injective function $f : A \rightarrow B$.

We will not develop the concept of cardinality $|A|$ for a while, so for now this symbol will only appear as a relation.

Theorem 3.1.5 (Cantor-Schröder-Bernstein). Suppose A, B are sets and there exists injective functions $f : A \rightarrow B$ and $g : B \rightarrow A$, then $A \approx B$.

Proof. First consider the sequence of sets $A_0 := A \setminus g(B)$ and for all $n > 0$, $A_n := (g \circ f)(A_{n-1})$. Then write $A^* := \bigcup_n A_n$ and $B^* = f(A^*)$, these clearly satisfy $g(B^*) = (g \circ f)(A^*) \subseteq A^*$.

I now claim $g(B \setminus B^*) = A \setminus A^*$. For all $a \in A \setminus A^*$, as $a \notin A_0$ there must exist a $b \in B$ with $g(b) = a$. If $b \in B^* = f(A^*)$, then $a = g(b) \in (g \circ f)(A^*) \subseteq A^*$ a contradiction, therefore $A \setminus A^* \subseteq g(B \setminus B^*)$. Now for all $b \in B$, if $g(b) \in A^*$, then as $g(b) \notin A_0$ there must exist some $n > 0$ with $g(b) \in A_n$ and thus $g(b) = (g \circ f)(a)$ for some $a \in A_{n-1}$, so $b = f(a)$ for some $a \in A^*$ and thus $b \in f(A^*) = B^*$.

g now gives a bijection $B \setminus B^* \rightarrow A \setminus A^*$, and as f is a bijection $A^* \rightarrow B^*$ we can form the bijection

$$k(a) = \begin{cases} f(a) & \text{if } a \in A^* \\ g^{-1}(a) & \text{if } a \in A \setminus A^* \end{cases} \quad \square$$

3.2 Zermelo-Fraenkel axioms

We will express these axioms in a first-order language with equality, which we call $\mathcal{L}_{\in}^=$, with a single extra 2-ary relation symbol $\in$.

Axiom 1 (Extensionality). $(\forall x)(\forall y)((x = y) \leftrightarrow (\forall z)((z \in x) \leftrightarrow (z \in y)))$

Axiom 2 (Empty set). $(\exists x)(\forall y)(y \notin x)$

Axiom 3 (Pairing). $(\forall x)(\forall y)(\exists z)(\forall w)((w \in z) \leftrightarrow ((w = x) \vee (w = y)))$

By extensionality we know we can write $\emptyset$ to mean the unique empty set in ZF. We can also start to form the natural numbers $0 = \emptyset$, $1 = \{\emptyset, \emptyset\} = \{\emptyset\}$, and $2 = \{\emptyset, \{\emptyset\}\}$, but we can't yet go any further.

Axiom 4 (Union). $(\forall A)(\exists B)(\forall x)((x \in B) \leftrightarrow (\exists z)((z \in A) \wedge (x \in z)))$

So if we take $A = \{x, y\}$ then $B = x \cup y$, so we can now form $3 = \{0, 1, 2\} = \{0, 1\} \cup \{2\}$, and thus we can form all the natural numbers.

We will want to adopt the abbreviation $A \subseteq B$ to mean $(\forall x)(x \in B \rightarrow x \in A)$.

Axiom 5 (Powerset). $(\forall A)(\exists B)(\forall z)((z \in B) \leftrightarrow (z \subseteq A))$

Axiom 6 (Specification). For all $\mathcal{L}_{\in}^=$ -formulas $P(x, y_1, \dots, y_r)$, for some finite number of variables r , there exists the axiom $(\forall A)(\forall y_1) \dots (\forall y_r)(\exists B)((x \in B) \leftrightarrow ((x \in A) \wedge P(x, y_1, \dots, y_r)))$.

This lets us form the intersection, let C be a nonempty set with $A \in C$, then

$$\bigcap C := \{x \in A \mid (\forall z)((z \in C) \rightarrow (x \in z))\}$$

and for sets A, B , we can form the product

$$A \times B := \{w \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid (\exists a)(\exists b)((a \in A) \wedge (b \in B) \wedge (w = \{\{a\}, \{a, b\}\}))\}$$

and the set of functions

$$B^A := \{w \in \mathcal{P}(A \times B) \mid (\forall a)((\exists x)(a \in x \wedge x \in w) \wedge (\forall y)(\forall z)((a \in y \wedge a \in z) \rightarrow y = z))\}$$

Definition 3.2.1. For all sets a , define the **successor** of a to be the set $a^{\dagger} := a \cup \{a\}$. A set A is **inductive** if $(\emptyset \in A) \wedge (\forall x)((x \in A) \rightarrow (x^{\dagger} \in A))$.

Axiom 7 (Infinity). There exists an inductive set.

Definition 3.2.2. We can use specification to form the set of inductive sets, and take their intersection, call this ω .

Proposition 3.2.3. $\mathbb{N}$ is an inductive set, and if B is an inductive set, then $\mathbb{N} \subseteq B$. If $A \subseteq \mathbb{N}$ is inductive, then $A = \mathbb{N}$.

This is proof by induction, if P is an $\mathcal{L}_{\in}^=$ -formula and we let $A = \{x \in \mathbb{N} \mid P(x)\}$, then if we prove a base case and an inductive step, we get $A = \mathbb{N}$.

3.3 Well orderings

Definition 3.3.1. A linear ordering $(A; \leq)$ is a **well-ordering** if every nonempty subset of A has a least element.

Definition 3.3.2. If two linear orders $\mathcal{A}_1 = (A_1; \leq_1)$ and $\mathcal{A}_2 = (A_2; \leq_2)$ are isomorphic we call them **similar** and write $\mathcal{A}_1 \simeq \mathcal{A}_2$. Furthermore, if $\beta : A_1 \rightarrow A_2$ is injective and $\forall a, b \in A_1$ we have $a \leq_1 b \implies \beta(a) \leq_2 \beta(b)$, we say β is **order-preserving**.

Definition 3.3.3. Given $\mathcal{A}_1, \mathcal{A}_2$ as above the **reverse lexicographic product** is the linear order $\mathcal{A}_1 \times \mathcal{A}_2 := (A_1 \times A_2, \leq)$ with the ordering $(a_1, a_2) \leq (b_1, b_2)$ iff either $a_2 \leq_2 b_2$ or $a_2 = b_2$ and $a_1 \leq_1 b_1$. Their **sum** is the linear order $(A_1 \sqcup A_2; \leq)$ where $\leq$ is the union of $\leq_1, \leq_2$, and $a_1 \leq a_2$ for all $a_1 \in A_1$ and $a_2 \in A_2$.

The key example is $\{0, 1\} \times \mathbb{N} \simeq \mathbb{N} \not\simeq \mathbb{N} + \mathbb{N} \simeq \mathbb{N} \times \{0, 1\} \not\simeq \mathbb{N} \times \mathbb{N}$. The proof that the sum and product actually satisfy the linear ordering axioms is left as an exercise.

Lemma 3.3.4. If $\mathcal{A}_1$ and $\mathcal{A}_2$ are well-orderings, then so are $\mathcal{A}_1 + \mathcal{A}_2$ and $\mathcal{A}_1 \times \mathcal{A}_2$.

3.4 Ordinals

Definition 3.4.1. A set X is called **transitive** if every element of X is also a subset of X .

A set α is called an **ordinal** if α is a transitive set and the relation $<$ on α given by $x < y \iff x \in y$ is a strict well ordering.

Lemma 3.4.2. If α is an ordinal, then so is $\alpha^\dagger$.

Proof. For transitivity, if $\beta \in \alpha^\dagger$ then either $\beta \in \alpha$ or $\beta = \alpha$, either way $\beta \subset \alpha^\dagger$. As the ordering $\in$ makes α the maximal element of $\alpha^\dagger$, this ordering is still strict, and by definition $\alpha \notin \alpha$, so the ordering remains strict. $\square$

So we can do induction to see that all the natural numbers are ordinals.

Lemma 3.4.3. If α is an ordinal and $\beta \in \alpha$, then β is an ordinal.

Proof. If $y \in x \in \beta \in \alpha$, then as $\in$ is an ordering, so transitive, we get $y \in \beta$, so β is transitive and inherits the ordering from α , thus is an ordinal. $\square$

Lemma 3.4.4. If α and β are ordinals, and $\alpha \subset \beta$, then $\alpha \in \beta$.

Proof. As $\beta \setminus \alpha$ is nonempty, we may choose a minimal element γ , I now claim $\gamma = \alpha$.

For all $x \in \gamma$, we have $x \in \beta$ and so by minimality $x \in \alpha$, thus $\gamma \subseteq \alpha$. Now suppose there exists some $y \in \alpha \setminus \gamma$ then in the total ordering on β we must have either $y = \gamma$ or $\gamma \in y$, either way $\gamma \in \alpha$ by transitivity. $\square$

Definition 3.4.5. If α and β are ordinals, write $\alpha < \beta$ for $\alpha \in \beta$.

Proposition 3.4.6. Suppose α , β , and γ are ordinals, then the following all hold:

1. if $\alpha < \beta$ and β, γ , then $\alpha < \gamma$;
2. if $\alpha \leq \beta$ and $\beta \leq \alpha$, then $\alpha = \beta$;
3. exactly one of $\alpha < \beta$, $\alpha = \beta$, or $\alpha > \beta$ holds;
4. if X is a nonempty set of ordinals, then X has a least element.

Proof. (1) is just by transitivity of γ .

(2) must hold, otherwise we would have $\alpha \in \beta \in \alpha$ a contradiction.

(3) We know at most one of these hold, and from the definition we can see $\alpha \cap \beta$ is also an ordinal. If neither hold, $\alpha \not\subseteq \beta$ and $\beta \not\subseteq \alpha$, then $\alpha \cap \beta \subset \alpha$ hence $\alpha \cap \beta \in \alpha$, by doing the same with β we get $\alpha \cap \beta \in \alpha \cap \beta$, which is a contradiction.

(4) Take $\delta \in X$, then the least element of X is the least element of $\delta \cap X \subseteq \delta$. $\square$

Corollary 3.4.7. If X is a nonempty set of ordinals, then $\bigcup X$ is an ordinal.

Proof. If $y \in z \in \bigcup X$, then there exists some $x \in X$ with $z \in x$, so $y \in x$, and thus $y \in \bigcup X$. $\square$

Corollary 3.4.8. ω is an ordinal.

Definition 3.4.9. If $(A; \leq)$ is a well-ordering, then we call $X \subset A$ an **initial segment** of A if whenever $y < x \in X$, then $y \in X$, X is proper if $X \neq A$.

Lemma 3.4.10. If $(A; \leq)$ is a well order, with X a proper initial segment, then there exists $x \in A$ such that $X = \{a \in A \mid a < x\}$.

Proof. Let x be the minimal element of $A \setminus X$, then for all $y < x$ we have $y \in X$. If $z \in X$, then $z \neq x$, so if $z > x$ we must have $x \in X$, a contradiction. $\square$

Lemma 3.4.11. If $(A; \leq)$ is a well-ordering, $f : A \rightarrow A$ is order-preserving, and $f(A)$ is an initial segment of A , then $f(a) = a$ for all $a \in A$.

Proof. Suppose this was not the case, then consider x , the least element of the set $\{y \in A \mid f(y) \neq y\}$, then f is the identity on the initial segment $A[x]$, and hence as f is injective and $f(x) \neq x$, we must have $f(x) > x$. But now $x \notin f[A]$, even though x is order preserving, a contradiction. $\square$

Corollary 3.4.12. If $\alpha \neq \beta$ are ordinals, then $\alpha \not\approx \beta$.

For the next proof we need the next ZF axiom.

Axiom 8 (Replacement). Suppose $F(x, y, z_1, \dots, z_r)$ is a property of sets expressible in $\mathcal{L}_\in^\equiv$ such that, whenever $s_1, \dots, s_r$ are sets, for every set a there exists a unique set b such that $F(a, b, s_1, \dots, s_r)$ holds. Then for any set A , we can form the set

$$B = \{b \mid (\exists a)((a \in A) \wedge F(a, b, s_1, \dots, s_r))\}$$

replacing each $a \in A$ with the unique b such that $F(a, b, s_1, \dots, s_r)$ holds.

Theorem 3.4.13. If $(A; \leq)$ is a well-ordering, then there exists a unique ordinal α which is similar to $(A; \leq)$.

Proof. First, uniqueness is exactly the previous corollary, as if α and β are both similar to $(A, \leq)$, then they are similar to each other, and hence equal.

For existence, consider the set $X = \{x \in A \mid A[x] \text{ is similar to an ordinal}\}$, then for each x there is a unique ordinal α_x such that $\alpha_x \simeq A[x]$, so we can use replacement to form the set $S = \{\alpha_x \mid x \in X\}$. As S is a set of ordinals, to show it is an ordinal itself, it suffices to show transitivity. If we have $\beta \in \alpha_x \in S$, with $\pi : A[x] \rightarrow \alpha_x$ the similarity, then we let $y = \pi^{-1}(\beta)$ which makes π a similar $A[y] \rightarrow \beta$, so $\beta = \alpha_y \in S$. This in fact says X is an initial segment of A . Suppose it is proper, then we must have $X = A[z]$ for some $z \in A$, but then $z \in X = A[z]$ a contradiction, so $X = A$ and the map $x \mapsto \alpha_x$ is a similarity between X and S . $\square$

3.5 Transfinite induction and recursion

Theorem 3.5.1 (Transfinite induction). Suppose $P(x)$ is a property of sets such that for all ordinals α , if $P(\beta)$ holds for all ordinals $\beta < \alpha$, then $P(\alpha)$ holds; then $P(\gamma)$ holds for all ordinals γ .

Proof. If $\alpha = 0$, then $P(\beta)$ holds vacuously, so $P(0)$ holds. Now consider the set of ordinals γ for which P doesn't hold, by assumption this can't have a minimum, and so must be empty. $\square$

Theorem 3.5.2. For all infinite ordinals α , we have $\alpha \approx \alpha \times \alpha$.

Proof. We already know the result holds if α is any countable ordinal, so we may assume α is uncountable, now by transfinite induction we just have to assume for all $\omega \leq \beta < \alpha$, that $\beta \approx \beta \times \beta$ and show $\alpha \approx \alpha \times \alpha$. We may also assume that if $\beta < \alpha$, then $\beta \not\approx \alpha$, from which it follows $\beta^\dagger \not\approx \alpha$. Finally, by Cantor-Schröder-Bernstein we just need to find an injective function $\alpha \times \alpha \rightarrow \alpha$.

First, suppose we have a well-ordering $\leq$ of $A = \alpha \times \alpha$ such that for all $x \in A$, $|A[x]| < |\alpha|$, then we may find an ordinal γ with $\gamma \simeq (A; \leq)$, and call the similarity f , then for all $\eta \in \gamma$ as f is a similarity, it gives a bijection $\eta \rightarrow A[f(\eta)]$, so $\eta \approx A[f(\eta)]$ which has cardinality strictly less than α , so by trichotomy we have $\eta < \alpha$ equivalently $\eta \in \alpha$, so $\gamma \subseteq \alpha$ and we are done.

Now, we must produce such a well-ordering on A . For all $\lambda < \alpha$ consider the set $A_\lambda := \{(\theta, \zeta) \mid \max(\theta, \zeta) = \lambda\}$ then we define $\leq$ on A by $(\theta, \zeta) \leq (\theta', \zeta')$ iff $\max(\theta, \zeta) < \max(\theta', \zeta')$ or if $\max(\theta, \zeta) = \max(\theta', \zeta')$ and either $\zeta < \zeta'$ or $\zeta = \zeta'$ and $\theta \leq \theta'$, it is left as an exercise to show this is a well-ordering on A . Now for $x = (\theta, \zeta) \in A$, if we take $\lambda = \max(\theta, \zeta)$, then $A[x] \subseteq A_{\lambda^\dagger} \subset \lambda^\dagger \times \lambda^\dagger$, and by our initial assumptions, we know $\lambda^\dagger \times \lambda^\dagger \approx \lambda$ which has cardinality strictly less than α . $\square$

Corollary 3.5.3. If $(A; \leq)$ is an infinite well-ordering, then $A \approx A \times A$.

Proof. We may find an ordinal α with $A \approx \alpha \approx \alpha \times \alpha \approx A \times A$. $\square$

Corollary 3.5.4. Assuming the axiom of choice, if A is any infinite set, then $A \approx A \times A$.

Proof. We will see that the axiom of choice lets us make any set well-ordered. $\square$

From the axioms of replacement, we have the notion of an operation F on sets, some property $F(x, y)$ of sets such that for each set a there is a unique set b such that $F(a, b)$ holds.

Theorem 3.5.5 (Transfinite recursion). If F is an operation on sets, then there is an operation G such that for all ordinals α , $G(\alpha) = F(G \upharpoonright \alpha)$, where we define $G \upharpoonright \alpha : \alpha \rightarrow \{G(\beta) \mid \beta < \alpha\}$. If G' is another operation such that $G'(\alpha) = F(G' \upharpoonright \alpha)$ for all ordinals α , then $G(\alpha) = G'(\alpha)$ for all ordinals α .

Proof. Non-examinable. $\square$

We can use this to prove a version of the Lindenbaum lemma which doesn't assume $\mathcal{L}$ is countable.

Proposition 3.5.6. Suppose $\mathcal{L}$ is a first-order language whose alphabet of symbols A is well-ordered, then if Σ is a closed set of $\mathcal{L}$ -formulas, we can find a complete consistent set $\Sigma^* \supseteq \Sigma$ of closed $\mathcal{L}$ -formulas.

Proof. As A is well-ordered, we can also well order the set of finite sequences in A which we call S . Hence we get a well-ordering on any subset of S , in particular the set of closed $\mathcal{L}$ -formulas. This set of closed $\mathcal{L}$ -formulas is now similar to some ordinal λ and index our set of $\mathcal{L}$ -formulas as $\{\phi_\alpha \mid \alpha < \lambda\}$.

Now, for each ordinal α , define the a consistem set $G(\alpha) \supseteq \Sigma$ of closed $\mathcal{L}$ -formulas by transfinite induction as follows:

$$G(\alpha) = \begin{cases} \Sigma \cup \bigcup_{\beta < \alpha} G(\beta) \cup \{\phi_\alpha\} & \text{if } \alpha < \lambda \text{ and } \Sigma \cup \bigcup_{\beta < \alpha} G(\beta) \vdash \phi_\alpha \\ \Sigma \cup \bigcup_{\beta < \alpha} G(\beta) \cup \{\neg\phi_\alpha\} & \text{if } \alpha < \lambda \text{ and } \Sigma \cup \bigcup_{\beta < \alpha} G(\beta) \not\vdash \phi_\alpha \\ \Sigma \cup \bigcup_{\beta < \lambda} G(\beta) & \text{if } \alpha \geq \lambda \end{cases}$$

now with the argument similar to the countable case, by with transfinite induction, we see for all $\beta < \alpha$, we have $G(\beta) \subseteq G(\alpha)$, and each $G(\alpha)$ is complete. Now if we set $\Sigma^* = G(\lambda)$, this is consistent and complete. $\square$

We can use similar arguments to give generalisations of the other parts of the Model existence theorem, which will eventually give us a Completeness and Compactness theorem for any $\mathcal{L}$ with a well-ordered alphabet.

Axiom 9 (Regularity). $(\forall x)((x \neq \emptyset) \rightarrow (\exists a)((a \in x) \wedge (a \cap x = \emptyset)))$

We have now stated all nine axioms of Zermelo-Fraenkel set theory.

3.6 Choice

Axiom 10 (Choice). Suppose A is a set of nonempty sets, then there exists a function $f : A \rightarrow \bigcup A$, such that for all $a \in A$, we have $f(a) \in a$.

If X is a non-empty set and $A = \mathcal{P}(X) \setminus \{\emptyset\}$, then AC gives us a function $f : A \rightarrow X$, such that for each $\emptyset \neq Y \subseteq X$, we have $f(Y) \in Y$.

If my set comes with a well-ordering, then by taking $f(Y)$ to be the minimum element of Y , I don't need the axiom of choice, this sort of goes the other way.

Theorem 3.6.1 (Hartog's lemma). For any set X , there exists an ordinal α such that there is no injective function $h : \alpha \rightarrow X$.

Proof. Consider the set $\mathcal{Y} = \{(Y; \leq_Y) \mid Y \subseteq A, \leq_Y \text{ is a well order on } Y\}$, then we use the axiom of replacement to form S by replacing each $(Y; \leq_Y)$ with an ordinal similar to it. Then S is in fact the set of ordinals which admit an injection into X , then let $\sigma = \bigcup S$, an ordinal, and now for all $\beta \in S$, we have $\sigma^\dagger > \beta$, so $\sigma^\dagger \notin S$. $\square$

Theorem 3.6.2. Suppose X is a non-empty set, and $f : \mathcal{P}(X) \setminus \{\emptyset\} \rightarrow X$ is a choice function, then there is a well-ordering $\leq$ on X .

Proof. Let q be some set with $q \notin X$, and consider $\tilde{X} = X \cup \{q\}$, using transfinite recursion, there is an operation G by for all ordinals γ :

$$G(\gamma) = \begin{cases} f(X \setminus \{G(\beta) \mid \beta < \gamma\}) & \text{if } X \setminus \{G(\beta) \mid \beta < \gamma\} \neq \emptyset \\ q & \text{otherwise} \end{cases}$$

if $q \notin \text{im}(G \upharpoonright \gamma)$ for some γ , then $G \upharpoonright \gamma$ is an injective function $\gamma \rightarrow X$, now by Hartog's lemma, there is some ordinal α with $G(\alpha) = q$, if we take the least such α , then $G \upharpoonright \alpha : \alpha \rightarrow X$ is a bijection, and this defines a well-ordering on X . $\square$

Axiom 11 (Well ordering principle). For all sets A , there is a well-ordering on A .

Corollary 3.6.3 (ZF). The well ordering principle is equivalent to the axiom of choice.

Proof. The axiom of choice gives a choice function of every set, which gives a well-ordering as above. If a set is well-ordered, then we may take the minimum element choice function we defined earlier. $\square$

Corollary 3.6.4 (ZFC). For all sets A , there exists an ordinal with $A \approx \alpha$.

Corollary 3.6.5 (ZFC). For all sets A and B , either $|A| \leq |B|$ or $|B| \leq |A|$.

Corollary 3.6.6 (ZFC). If A is any infinite set, then $A \times A \approx A$.

Lemma 3.6.7 (ZFC). Suppose A and B are non-empty sets, then there is an injective function $A \rightarrow B$ iff there is a surjection $B \rightarrow A$.

3.7 Cardinals

Definition 3.7.1. An ordinal α is called a **cardinal** if for all $\beta < \alpha$, $\alpha \not\approx \beta$.

From the problem sheets we know all finite ordinals and ω are cardinals, if ω is an infinite ordinal, then $\omega^\dagger$ isn't a cardinal, the set of countable ordinals is a cardinal, and every set A is equinumerous to a unique cardinal α .

Definition 3.7.2. The α above is called the **cardinality** of A , and we write $|A| = \alpha$.

So if we accept the axiom of choice, we get a much shorter proof of Cantor-Schröder-Berstein just by comparing cardinalities.

Definition 3.7.3. We define the sequence of **aleph** cardinals by transfinite recursion as follows. For all ordinals α , $\aleph_\alpha$ is a cardinal. $\aleph_0 = \omega$, and $\aleph_\alpha$ is the least cardinal which is greater than $\aleph_\beta$ for all ordinal $\beta < \alpha$. The existence of these is guaranteed by Cantor's theorem.

Definition 3.7.4 (Cardinal arithmetic). If A and B are disjoint sets with cardinalities $|A| = \kappa$ and $|B| = \lambda$, then define $\kappa + \lambda = |A \cup B|$, $\kappa \cdot \lambda = |A \times B|$, and $\kappa^\lambda = |A^B|$.

These cardinalities don't depend on the choice of A and B .

Theorem 3.7.5. If $\kappa \leq \lambda$ are cardinals with λ infinite, then $\kappa + \lambda = \kappa \cdot \lambda = \lambda$. If $2 \leq \kappa \leq \lambda$ then $2^\lambda = \kappa^\lambda$.

Proof. For the first statement, as $\kappa \leq \lambda$, we have $\kappa \cdot \lambda = |\kappa \times \lambda| \leq |\lambda \times \lambda| = |\lambda| = \lambda$, and we can easily find an injection $\lambda \rightarrow \kappa \times \lambda$. We also have $\lambda \leq \kappa + \lambda \leq \lambda + \lambda \approx 2 \cdot \lambda \approx \lambda$. The second part is left as an exercise. $\square$

Theorem 3.7.6. Suppose A is an infinite set of cardinality λ , and each element of A has cardinality at most κ , then $|\bigcup A| \leq \lambda \cdot \kappa$.

Proof. For each $a \in A$, the set S_a of surjective functions $\kappa \rightarrow a$ is non-empty, so by AC there is a function $F : A \rightarrow \bigcup S_a$ with $F(a) \in S_a$. Let $h : \lambda \rightarrow A$ be a bijection, then define $G : \lambda \times \kappa \rightarrow \bigcup A$ by $G(\alpha, \beta) = F(h(\alpha))(\beta)$, this is surjective and thus $|\bigcup A| \leq \lambda \cdot \kappa$. $\square$

From this we can deduce that for any infinite set A , the set of finite sequences of elements of A has the same cardinality. And, if B is a $\mathbb{Q}$ -basis for $\mathbb{R}$, then $|B| = |\mathbb{R}|$.

3.8 Zorn's lemma

Definition 3.8.1. A **chain** is a subset of a partially ordered set which is linearly ordered.

Axiom 12 (Zorn's lemma). If $(A; \leq)$ is a non-empty partially ordered set, in which every chain has a upper bound, then A has a maximal element.

Theorem 3.8.2. Assuming ZF, Zorn's lemma is equivalent to the axiom of choice.

Proof. (AC $\Rightarrow$ ZL) Given a poset $(A; \leq)$ satisfying the hypothesis of Zorn's lemma, let $f : \mathcal{P}(A) \setminus \{\emptyset\} \rightarrow A$ be a choice function, and suppose, for a contradiction, that A has no maximal element. Let $C \subseteq A$ be a chain, then by assumption there exists some $y \in A$ with $C \leq y$, and so there exists some $z \in A$ with $z > y$, so $C \subsetneq C \cup \{z\}$ is another chain, and we now use transfinite recursion to define an operation G such that for all ordinals $\beta < \alpha$, $G(\alpha) \in A$ and $G(\beta) < G(\alpha)$. For this we let

$$G(\alpha) = f(\{z \in A \mid z > G(\beta) \text{ for all } \beta < \alpha\})$$

which gives for every α an injection $G \upharpoonright \alpha : \alpha \rightarrow A$, which contradicts Hartog's lemma.

The other direction of the proof is left as an exercise. □